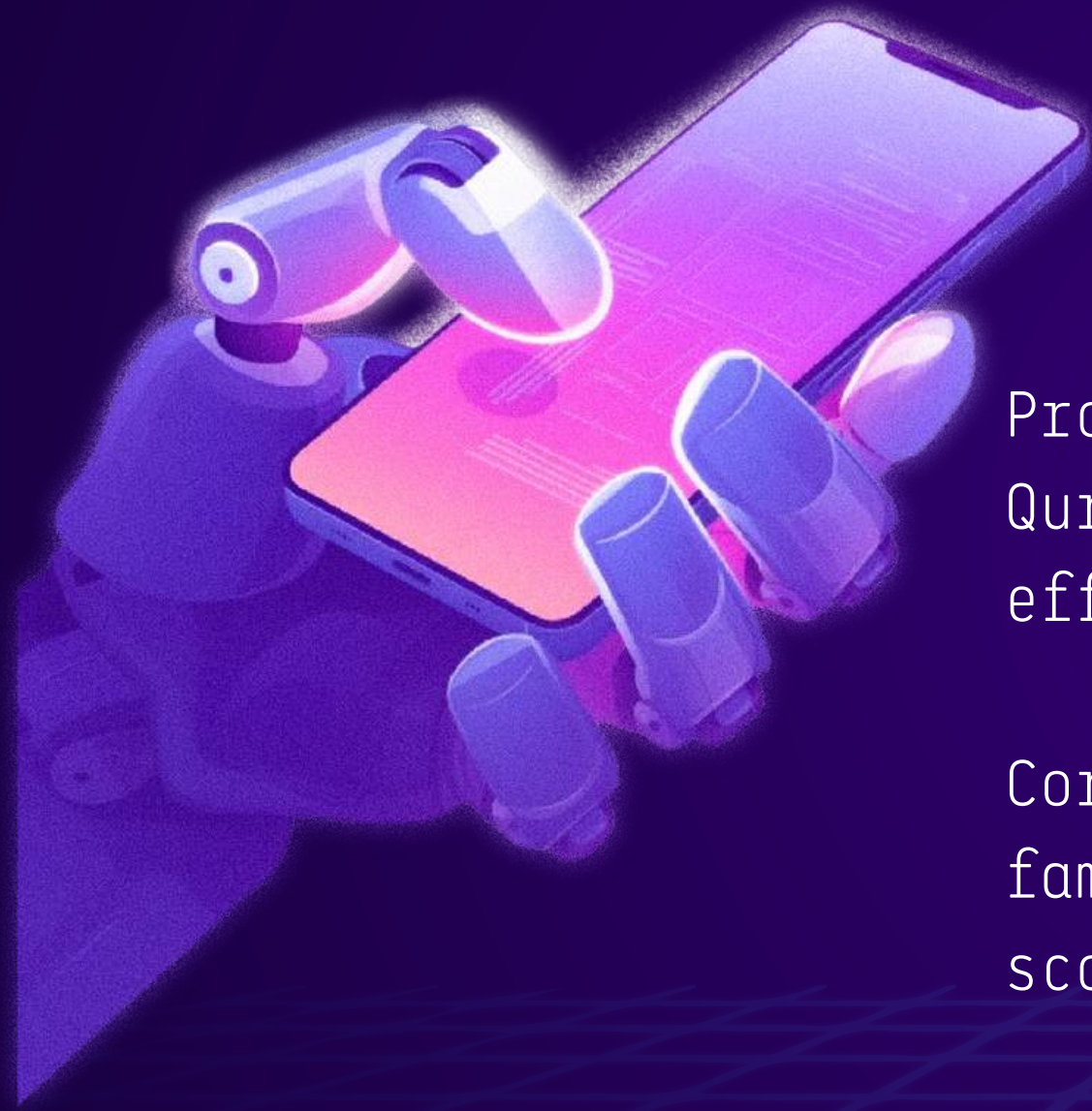


SMART CHARITY MANAGEMENT SYSTEM



Prepared by :

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- Nadine Ali Mohamed



SMART

RESOURCE ALLOCATION SYSTEM

Project Concept: Building an intelligent platform for Al-Qurain Al-Gedida Association to manage and distribute aid efficiently.

Core Models: Applying Classification to identify deserving families and Regression to calculate precise priority scores.

Main Goal: Transforming traditional manual distribution into a fair, transparent, and data-driven process for 1000+ individuals.

PROJECT PILLARS & OBJECTIVES

Core pillars designed to transform raw family data into actionable insights and fair distribution for Al-Qurain Al-Gedida Association.

Pilar 1: Data Collection

Pilar 3 : Smart Allocation

Pilar2:Priority Analysis

Pilar 4: Prediction & Improvement



KEY STEPS IN MACHINE LEARNING PROJECTS

01 Problem Definition

Defining the Problem of fair aid distribution and target families for El_Quran Association

02 Data Management

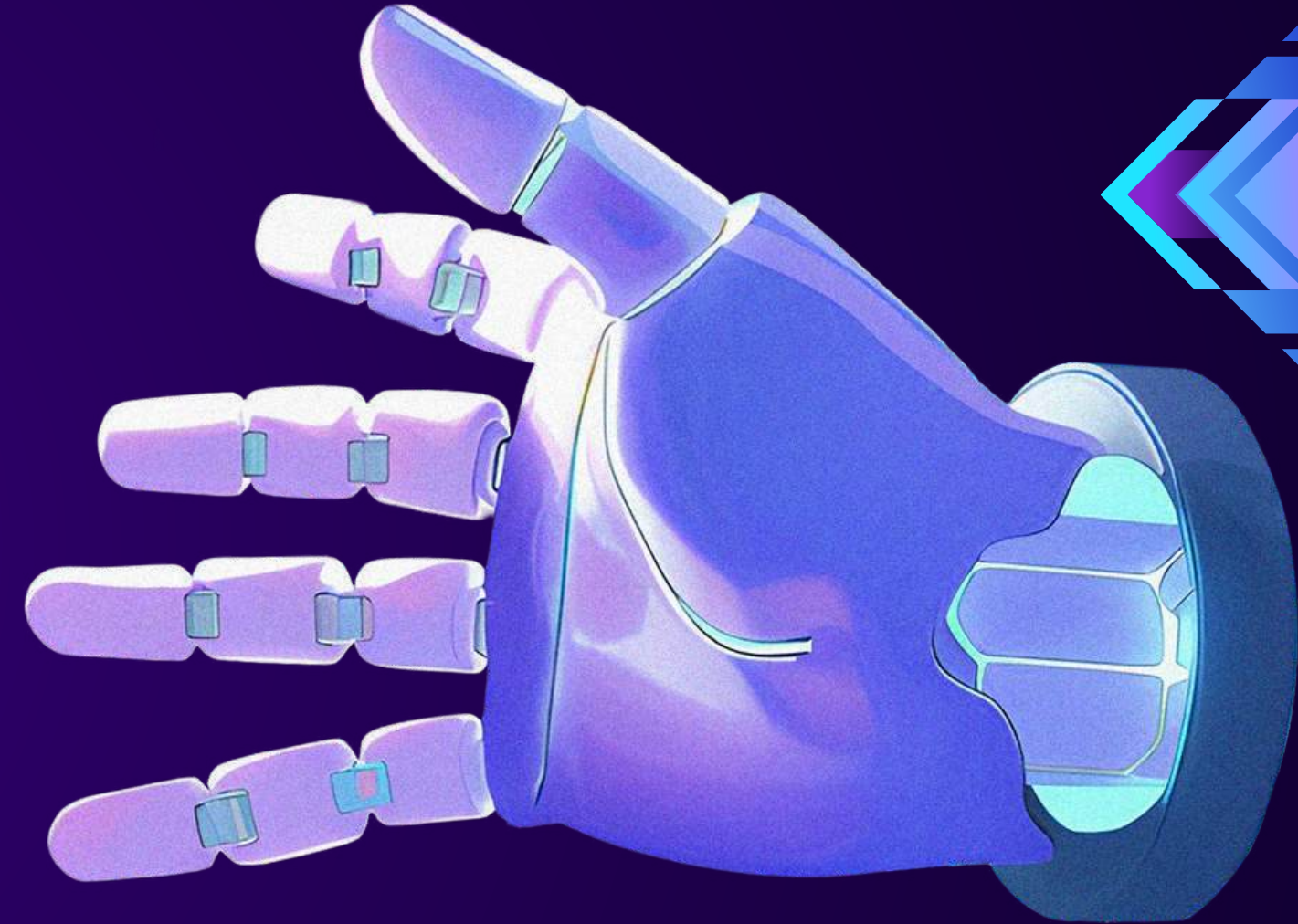
Collecting, processing, and cleaning family data to prepare it for analysis and prediction.

03 Model Development

Building and training {Regression & Random Forest} models to calculate priority scores.

04 Deployment & Monitoring

Deploying the system to distribute resources accurately and quickly based on real data.



IMPACT & COMMUNITY VALUE

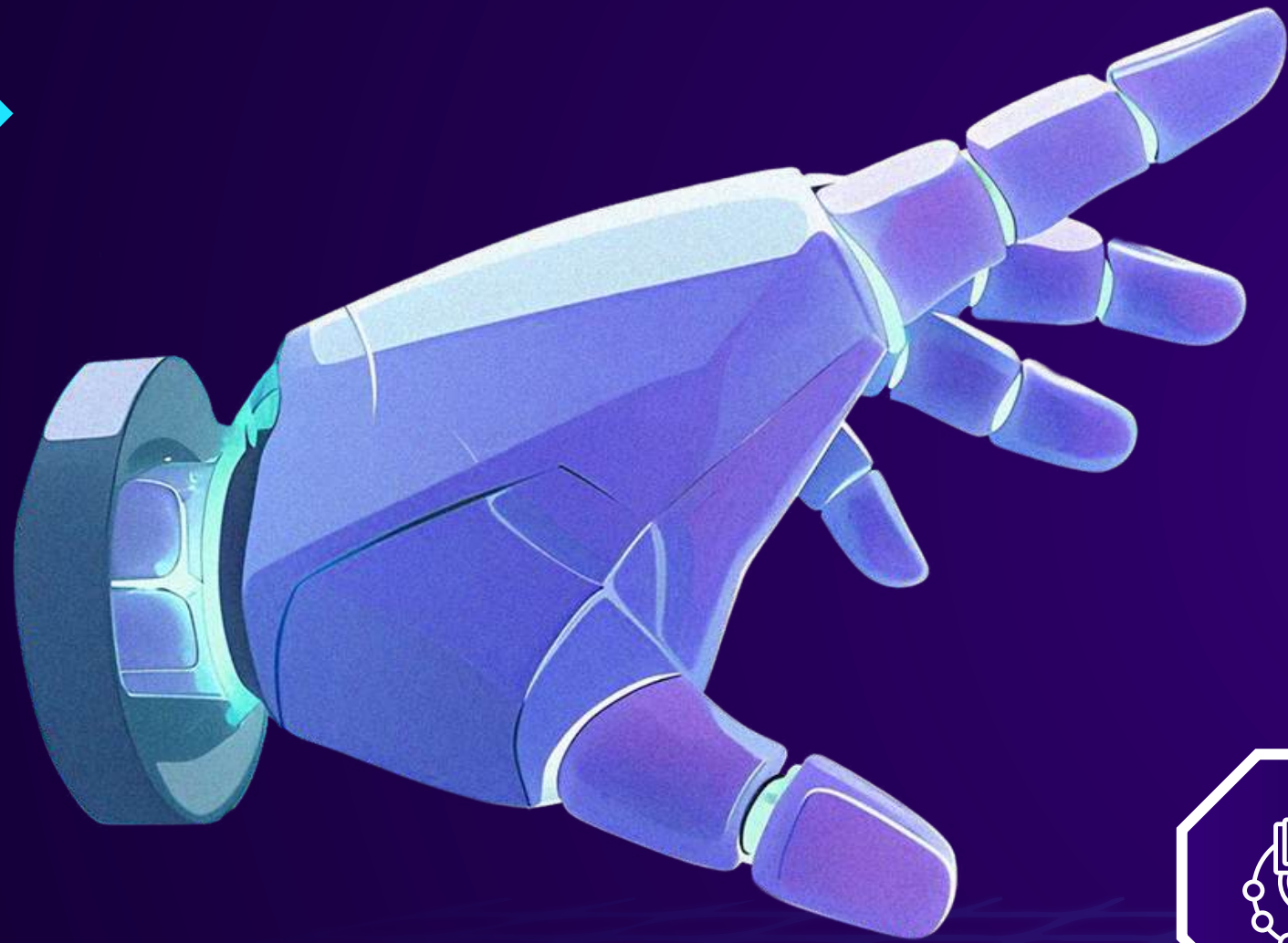
Fair Income Distribution: Ensuring charitable support reaches the most needy families in Al-Qurain Al-Jadida Association based on accurate priority scores.

Allocating the Designated Amount: The system calculates and determines the precise amount due for each family based on their status and actual degree of need.

Efficient Resource Management: Optimizing the budget usage to its fullest potential and preventing any waste of assistance.

Digital Transformation: Eliminating slow paperwork and manual work, and relying on a smart, fast system.





PROJECT CORE MODELS

Implementing supervised learning models to accurately evaluate family needs, predict priorities, and optimize resource allocation



Classification



Regression



Data Optimization

PROJECT KEY OUTCOMES

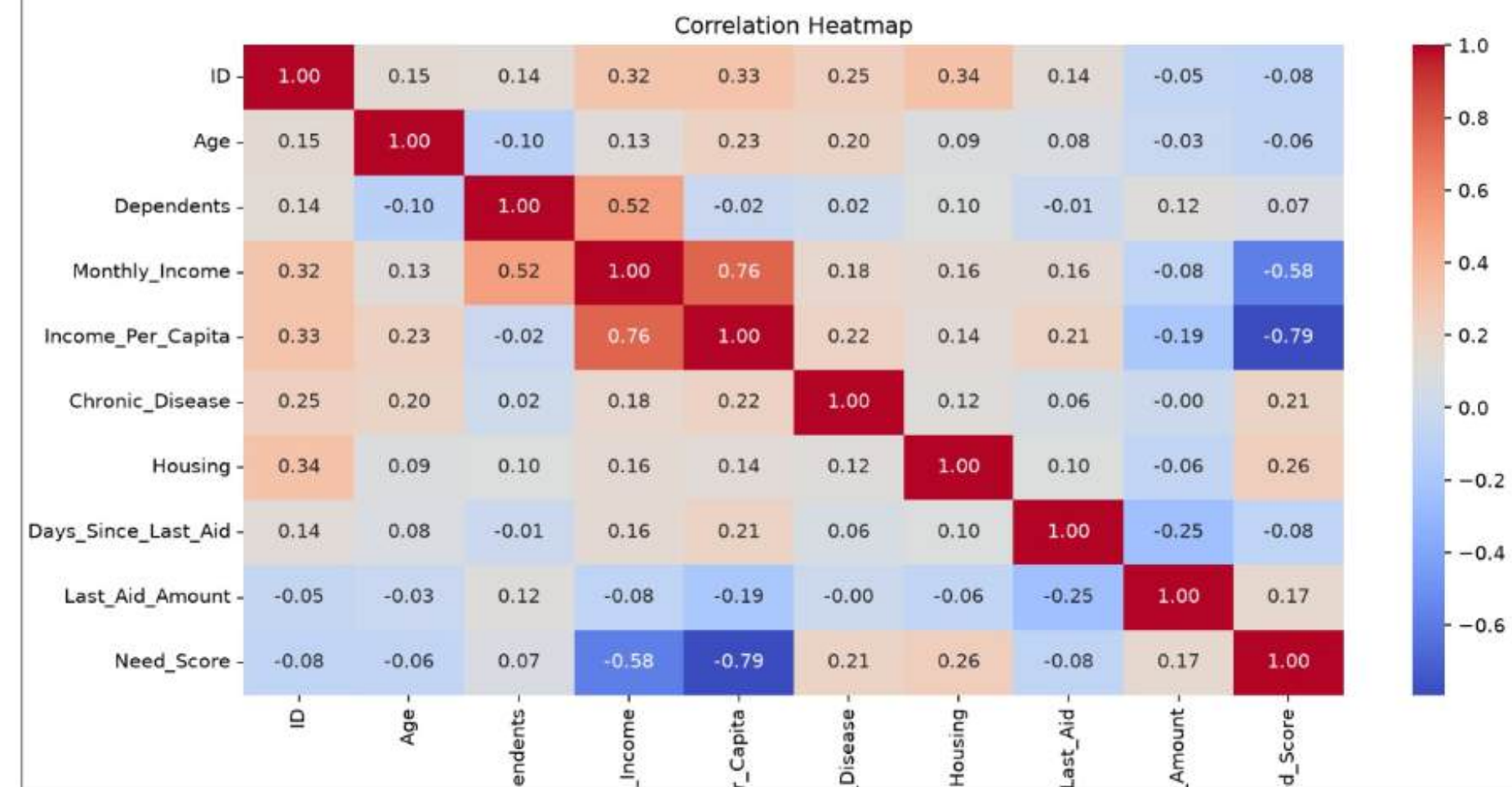
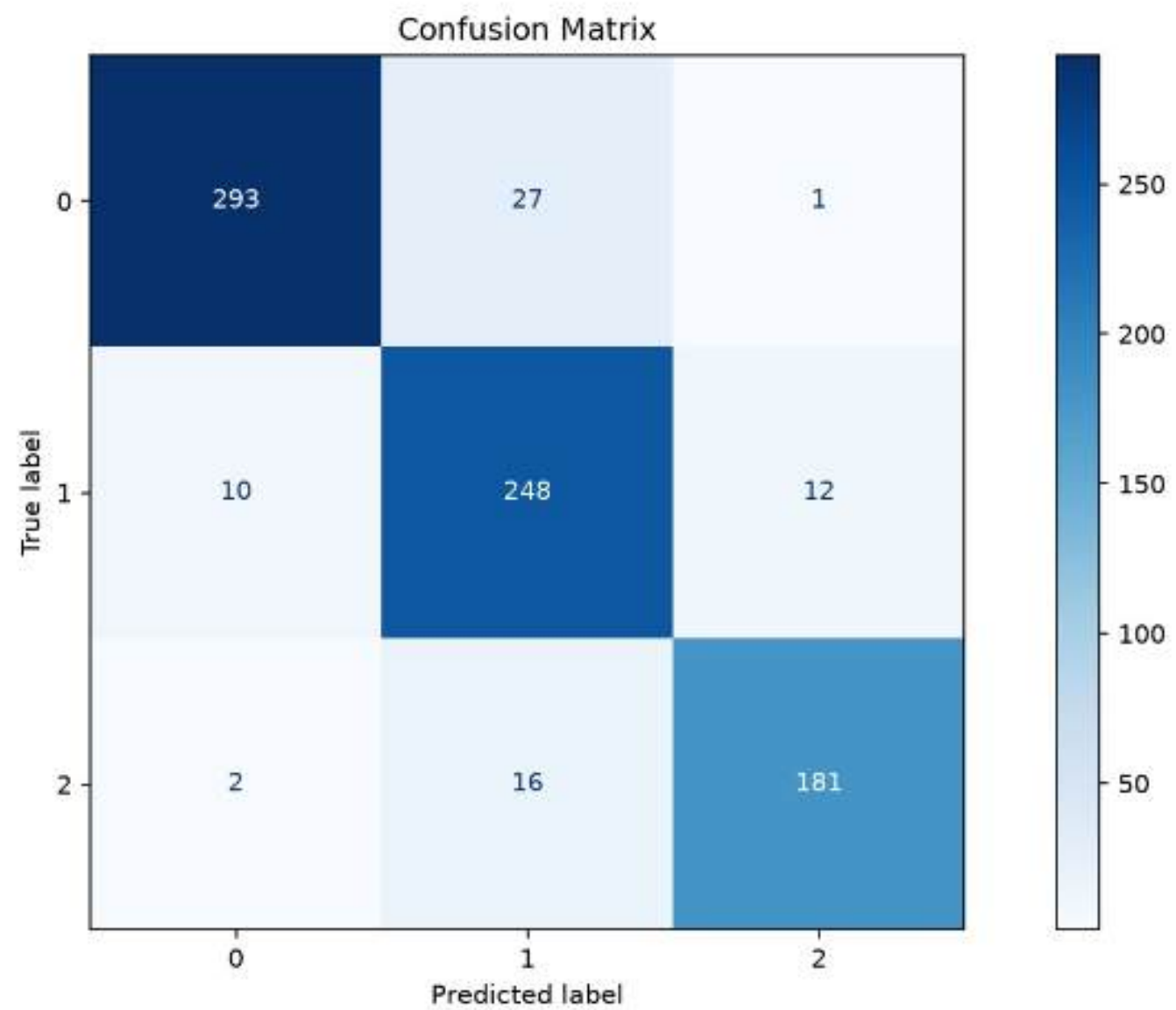
Smart Practical Application: Transforming the process of evaluating families and supporting them in the "Al-Qurain Al-Jadida Association" from a traditional framework to an integrated automated system.

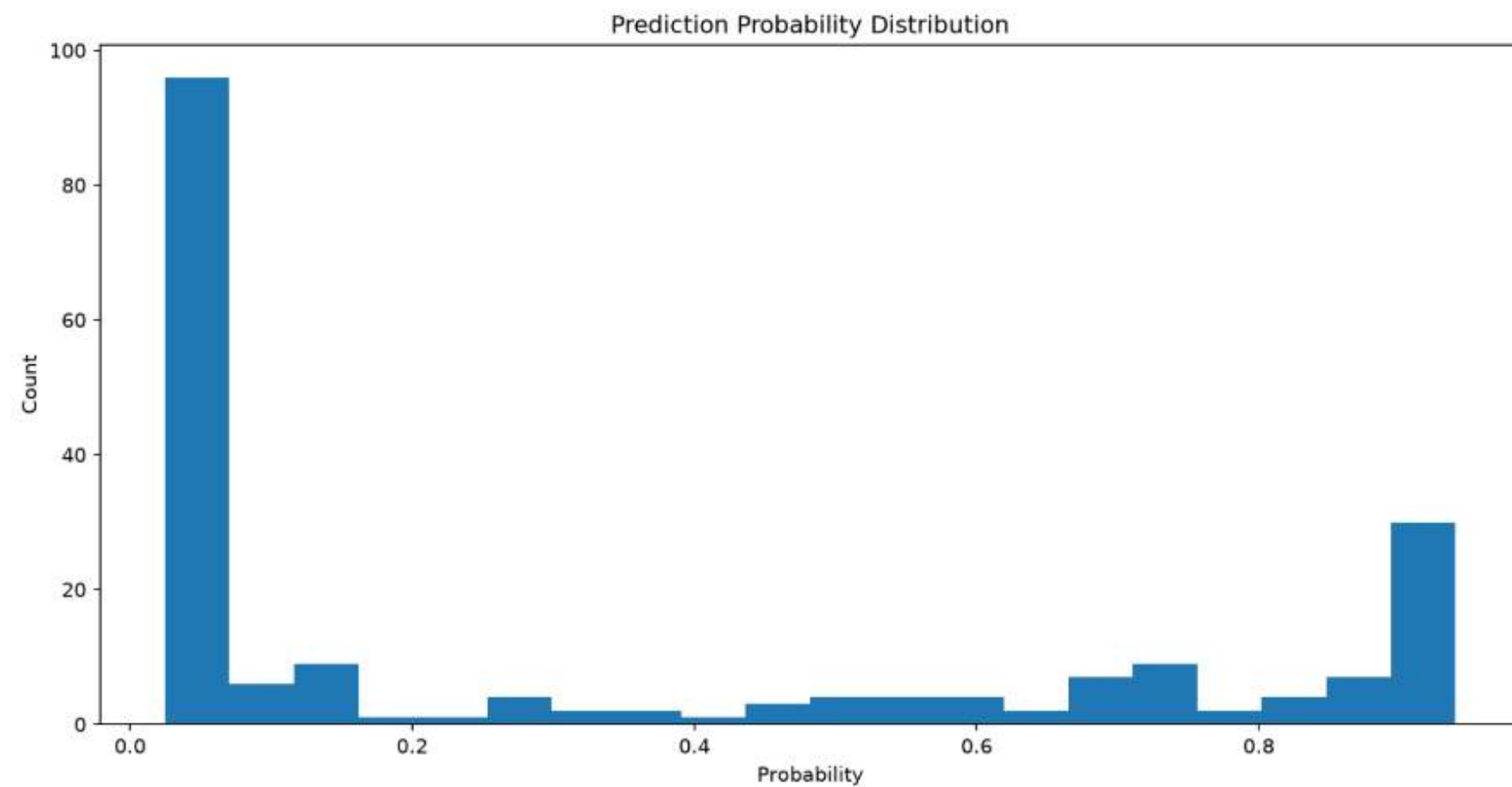
Precise Resource Allocation: Directing financial support within the one-million-pound budget to those who genuinely deserve it with utmost transparency.

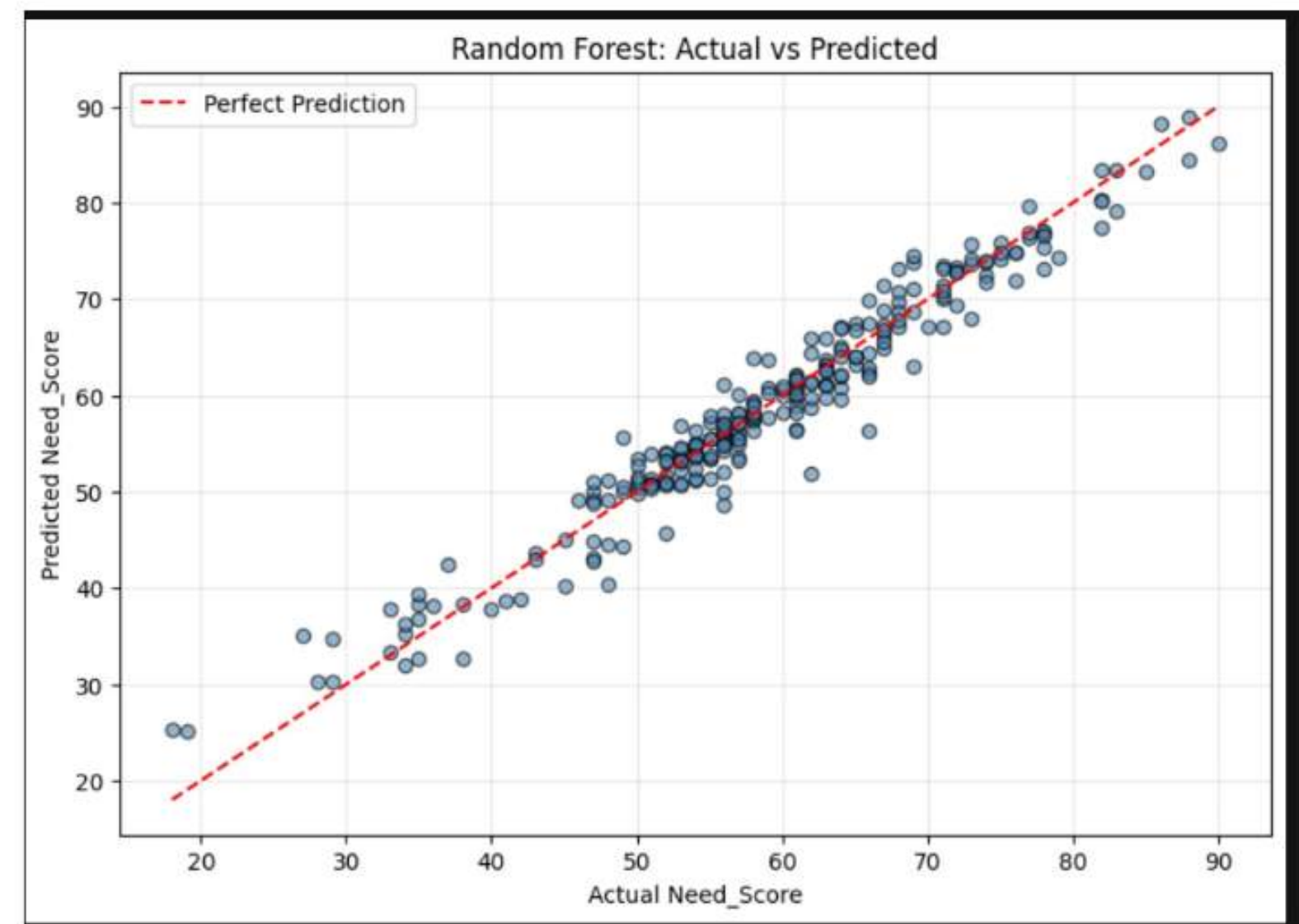
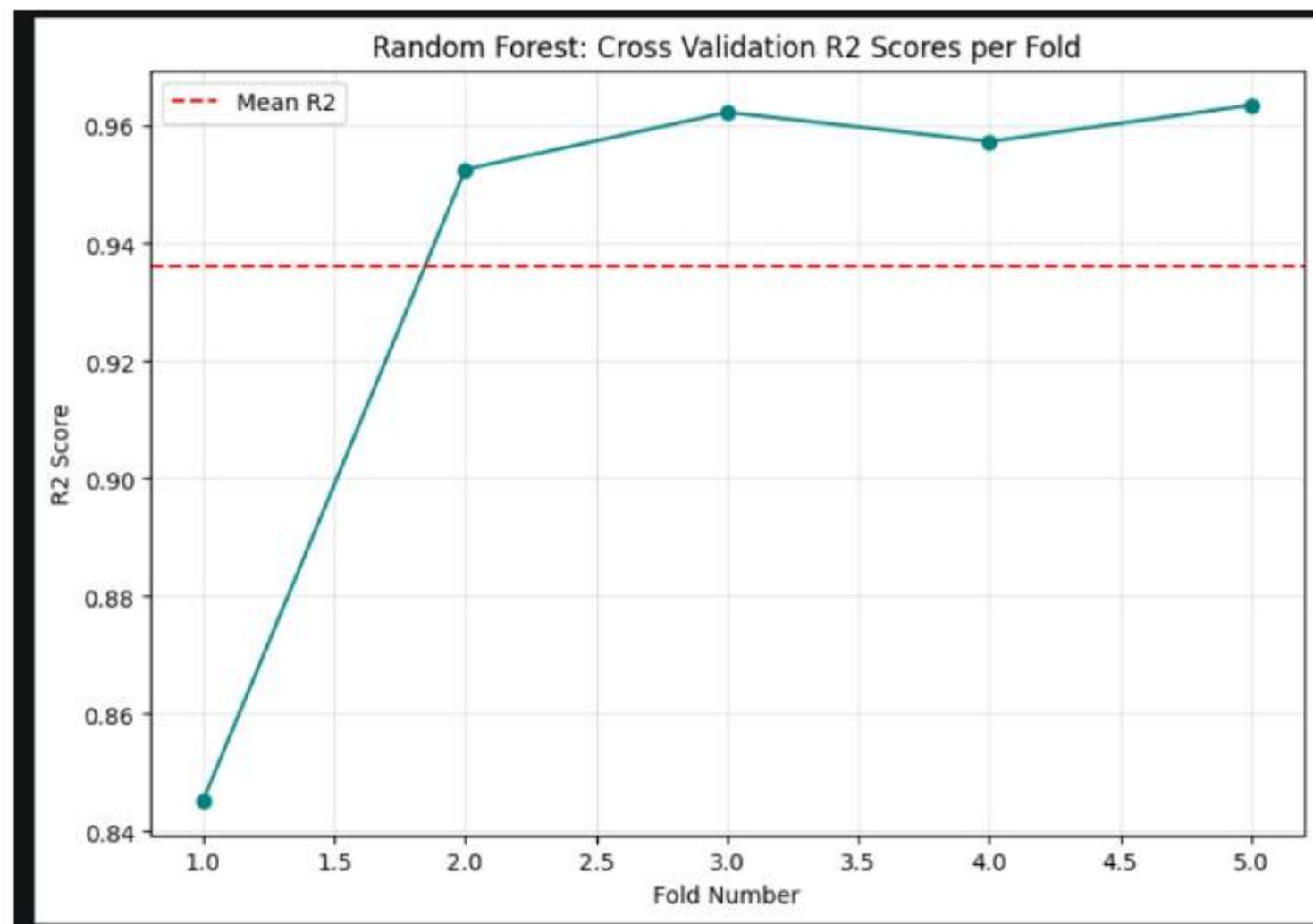
Saving Time and Effort: Eliminating human delays and relying on fast algorithms for decision-making.

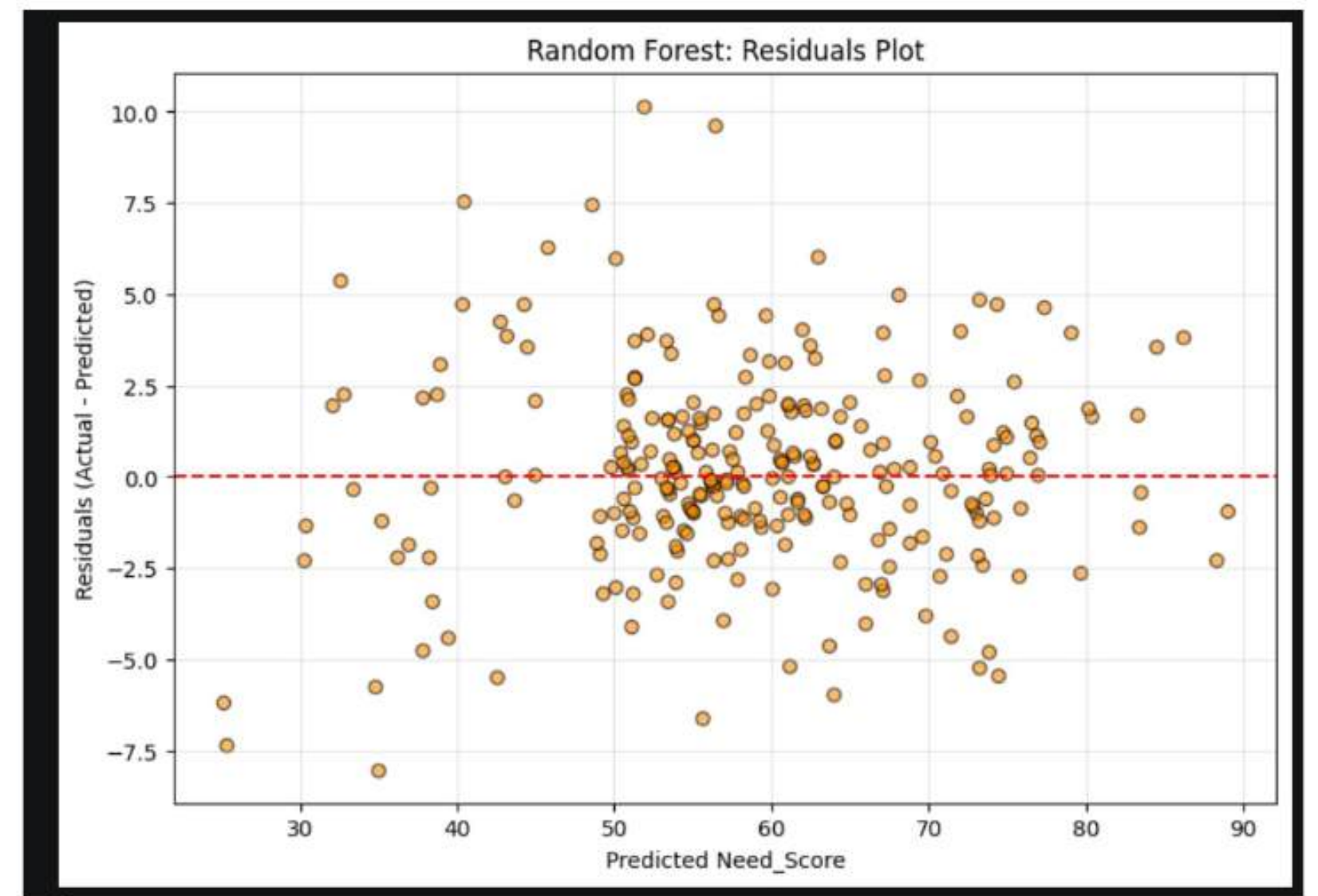
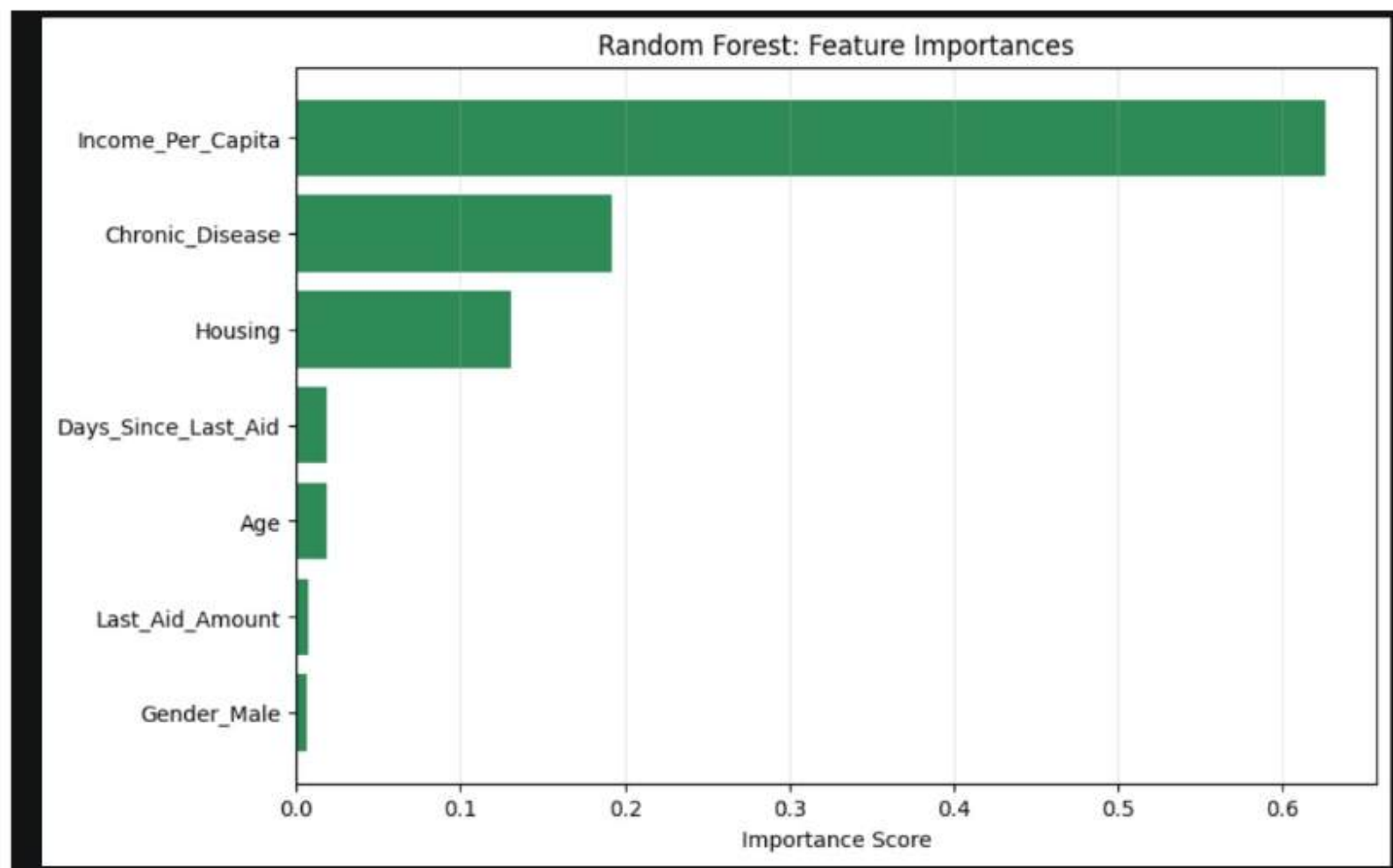
=== Models Accuracy Comparison ===

Model	MSE	MAE	RMSE	R2 Score	Mean CV R2
XGBoost	3.695960	1.506167	1.922488	0.986455	0.982866
SVR	4.198948	1.486916	2.049133	0.984611	0.984276
Random Forest	6.822774	2.036404	2.612044	0.974995	0.972978
AdaBoost	11.425260	2.661803	3.380127	0.958128	0.931719
Decision Tree	13.290322	2.676054	3.645589	0.951292	0.948775
Linear Regression	29.108204	3.713670	5.395202	0.893321	0.907612











THANKS